

Alice Gao

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Areas of interest

Human-computer interaction • UI Understanding • Computational Interfaces • Human-AI interaction • Data-driven design tools

Education

2023-present	PhD in Computer Science & Engineering, University of Washington
2021-2023	MSc in Computer Science & Engineering, University of Washington
2017-2021	BS in Computer Science, Princeton University

Publications

* denotes equal contribution

CONFERENCE AND JOURNAL PAPERS

- | | |
|------|---|
| 2026 | Alice Gao , Andrew N. Meltzoff, Maarten Sap, Katharina Reinecke. Framing an AI with Values Reduces AI Reliance in AI-supported Writing Tasks. <i>ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)</i> . Montreal, QC, Canada. June 2026. doi.org/10.1145/3805689.3812369

Nino Migineishvili, Alice Gao , Adinawa Adjabodjou, Dhanaraj Thaku, René Just, Katharina Reinecke. Regulating AI: Where U.S. State Policy and HCI (Mis)align. <i>ACM Conference on Human Factors in Computing Systems (CHI '26)</i> . Barcelona, Spain. April 2026. doi.org/10.1145/3772318.3791424 . |
| 2025 | Alice Gao *, Samyukta Jayakumar*, Marcello Maniglia*, Brian Curless, Ira Kemelmacher-Shlizerman, Aaron R. Seitz and Steven M. Seitz. Don't Look at the Camera: Achieving Perceived Eye Contact. <i>Journal of Vision</i> 2025; 25(11):8, doi.org/10.1167/jov.24.10.1094 . |
| 2024 | Alice Gao , Wataru Akahori, Naomi Yamashita, and Katharina Reinecke. Using Slack in the US and Japan: Surfacing Cultural Asymmetries in Overcoming Shortcomings. <i>In Submission</i> . |

WORKSHOP PAPERS & POSTERS

- | | |
|------|---|
| 2022 | K.J. Kevin Feng*, Alice Gao *, Johanna Suvi Karrass*. Towards Semantically Aware Word Cloud Shape Generation. <i>Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)</i> . Bend, OR. October 2022. |
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PREPRINTS

2026

Alice Gao, Weixi Tong, Rishab Vempati, Katharina Reinecke, R. Benjamin Shapiro, Tianyi Zhang, Jason Wu. Training Computer Use Agents to Assess the Usability of Graphical User Interfaces. April 2026. *In Submission*. [arXiv preprint](#).

Donghoon Shin, **Alice Gao**, Rock Pang, Katharina Reinecke, Emily Tseng. Interrogating Design Homogenization in Web Vibe Coding. March 2026. *In Submission*. [arXiv preprint](#).

Research Experience

Jan 2024 -
Present

Research Assistant, **Wildlab, University of Washington**

Advisors: Katharina Reinecke and R. Benjamin Shapiro

- Developed novel machine-learning algorithm to train computer use agents (CUAs) to automatically perform automated usability testing.
- Developing data-driven, machine learning models to learn user preferences of different demographics and automate generate of better, personalized UI designs.
- Developing semi-automated data-driven tools to support UI/UX designers in creating website design for broader audiences.
- Designed a new interventions to reduce AI overreliance during AI-supported writing by up to 30%.

Sep 2021 -
Jan 2024

Research Assistant, **GRAIL, University of Washington**

Advisors: Steve Seitz, Brian Curless, Ira Kemelmacher-Shlizerman

Developed machine learning techniques to create a model to redirect attention and eye gaze for a more attentive video conferencing experience.

Sep 2019 -
Feb 2021

Undergraduate Research Assistant, **Princeton Vision & Learning Lab**, Princeton University

Advisor: Jia Deng; mentored by Hei Law

Improved associative embedding predictions and grouping from a neural network to provide faster and more accurate predictions for a one-stage object detector in instance segmentation.

Teaching

University of Washington (graduate)

Spring 2026

CSE446: Machine Learning, *Teaching Assistant*

Fall 2025

CSE373: Data Structures & Algorithms, *Teaching Assistant*

Spring 2024

CSE581: Computer Ethics, *Teaching Assistant*

Fall 2024

CSEP557: Current Trends In Computer Graphics, *Teaching Assistant*

Winter 2024

CSE581: Computer Ethics, *Teaching Assistant*

Princeton University (undergraduate)

Spring 2020

COS426: Computer Graphics, *Undergraduate Teaching Assistant*

Fall 2018-Fall 2019

COS226: Algorithms & Data Structures, *Course Grader*

Industry Experience

Jun-Aug 2020

Software Engineering Intern in Online Account Opening; Capital One; New York City, New York
Created new endpoint and unit tests, laid framework for streaming data platform, and launched

new security groups.

- Jun-Aug 2019 *AI Design Lab Intern; Tezign; Shanghai, China*
Assisted in optimizing image retrieval to match similar designs and contributed to training a model in judging strong vs weak graphic designs.
- Jun-Aug 2018 *Technical and App Development Intern; Princeton Satellite Systems; Princeton, New Jersey*
Devised beginnings of gameplay for spacecraft simulation iOS app. Created 3D models for Phase II version of Direct Fusion Drive (DFD), a direct-drive, fusion-powered rocket engine.

Service and Leadership

- May 2018 -
Mar 2020 *co-President, Princeton Chinese Students Association*
Started a new Red Envelope fundraiser, acted as a liaison between guest speakers and CSA, participated in creating the first Asian+Students Council, organized and raised funding to host Steven Lim as a guest speaker, started weekly community nights (mahjong nights) that persist until today.

Skills

Languages: Python, R, JavaScript, SQL, TypeScript, C, Go
Frameworks: PyTorch, scikit-learn, Pandas, React, Sveltekit, Node.js, Django/DRF, Flask
Tools: Docker, Apptainer, Figma
Other: Machine Learning, Quantitative and qualitative research methods, Statistical modeling, Experimental design, Semi-structured Interviews

Last updated: August 12, 2026